

Automated Review Rating System

1 Deep Learning Research Findings

Objective: Explore whether deep learning models can outperform traditional machine learning for text classification (review rating prediction).

Key Observations:

1. **Model Selection:**
 - Traditional ML: Logistic Regression, Random Forest, SVM
 - Deep Learning: LSTM, GRU, or simple Dense networks
 - Deep learning models perform better on **large datasets** because they can capture sequential patterns and contextual relationships in text.
2. **Advantages of Deep Learning over Traditional ML:**
 - Handles complex patterns in text automatically (e.g., word order, context).
 - Can incorporate embeddings like Word2Vec or GloVe for semantic understanding.
 - Potentially higher accuracy on imbalanced datasets if trained properly.
3. **Limitations Observed:**
 - Requires **more computational resources** and longer training time.
 - Sensitive to hyperparameter choices and overfitting.
 - Smaller datasets may still perform better with classical ML.
4. **Finding Summary:**
 - For small-to-medium datasets, traditional ML with TF-IDF often gives competitive performance.
 - Deep learning is beneficial when dataset size grows or semantic understanding is critical.

2 UI Design Decisions

Goal: Build a simple, user-friendly interface for comparing predictions from two trained models.

Key Decisions:

1. **Single Text Area Input:**
 - Allows users to enter a review of any length.
 - Provides flexibility for testing real-world examples.

2. **Predict Button:**

- Explicit action triggers model prediction, avoiding unnecessary computations on each keystroke.

3. **Two-Column Layout:**

- Model A (Balanced) and Model B (Imbalanced) results displayed side by side.
- Facilitates **direct comparison** between models.

4. **Error Handling and Warnings:**

- Checks for empty input before prediction.
- Catches errors in model prediction (e.g., missing files, incompatible input).

5. **Page Configuration:**

- Wide layout to utilize screen space.
- Clear, descriptive title with a star icon for visual appeal.

3 User Flow

1. User navigates to the Streamlit app.
2. Enters a product review in the text area.
3. Clicks **Predict**.
4. The app:
 - Transforms the text using the corresponding vectorizers.
 - Feeds transformed text to each model.
5. Displays predictions in two columns:
 - **Left:** Model A prediction (Balanced dataset)
 - **Right:** Model B prediction (Imbalanced dataset)
6. User can repeat steps for different reviews.